SciCore-Mol Technical Report

Yuxuan Chen[1], Changwei Lv[2], Yunduo Xiao[2], Yukun Yan[2], Zheni Zeng*[3], and Zhiyuan Liu[2]

[1]School of Electronic and Computer Engineering, Peking University, Shenzhen, China
[2]Tsinghua University, Beijing, China
[3]School of Intelligence Science and Technology, Nanjing University, Nanjing, China
*Corresponding author: zengzn@nju.edu.cn

SciCore-Mol Code   HF SciCore-Mol Model

Abstract

Large language models (LLMs) show strong reasoning ability, but directly applying them as universal molecular solvers remains inefficient and unreliable. Molecular computation requires handling graph topology, stereochemistry, 3D geometry, structure-conditioned generation, and reaction-level quantitative prediction, whereas text-only LLMs operate over discrete symbolic sequences. Although SMILES-like molecular strings make molecules compatible with token-based language modeling, they compress molecular topology and geometry into one-dimensional sequences and do not explicitly expose the invariances and numerical structures required for chemical reasoning. Existing chemistry LLMs and tool-augmented systems improve usability, but many still treat the LLM mainly as a language interface, planner, decoder, or conditioning signal, leaving the core scientific computation to external task-specific modules. We present **SciCore-Mol**, a technical framework that augments an 8B-parameter LLM with three pluggable specialist modules: (1) a GVP-based *Topological Perception Module* for SE(3)-equivariant 3D encoding, (2) a latent-diffusion *Molecular Generation Module* for valid structure synthesis and editing, and (3) a Transformer-based *Reaction Sensing Module* for product prediction, yield estimation, and retrosynthesis. Modules are invoked by the LLM through special tokens and can be independently enabled or disabled, enabling a composable inference interface. Training follows a progressive three-stage recipe: (i) independent component pre-training with KL-regularized continual pre-training on a 400M-token chemistry corpus, (ii) cross-modal alignment, and (iii) joint multi-task fine-tuning on 300K instruction pairs. Experiments on SMolInstruct, MMLU-Chemistry, ChemBench4K, ORD, and MoleculeNet demonstrate improved molecular generation validity, reaction reasoning quality, and property prediction accuracy while maintaining general instruction-following capability.

Contents

1 Introduction

Large language models (LLMs) represent the current frontier of AI reasoning and instruction following [8; 26]. However, directly applying LLMs as universal scientific solvers remains inefficient and unreliable for structure-sensitive domains such as molecular science. Molecules are not merely linguistic sequences, but structured objects whose properties depend on atomic connectivity, stereochemistry, conformational geometry, and inter-molecular interactions. Although SMILES-like molecular strings make molecules compatible with text tokenizers, they serialize molecular graphs into one-dimensional token sequences and do not explicitly expose the topological and geometric invariances that are central to chemical reasoning [24; 1; 9; 15; 12; 30]. As a result, a pure LLM must implicitly recover graph connectivity, spatial relations, and chemical constraints from token co-occurrence, which is data-inefficient and prone to spurious correlations. Recent chemistry benchmarks further show that even strong LLMs can still fail on basic chemical reasoning tasks and produce unreliable or overconfident predictions, suggesting that scaling text-only LLMs alone is insufficient for robust molecular reasoning [17].

Several directions have been proposed to bridge this gap. Text-centric chemistry LLMs, such as MolT5, ChemLLM, BioT5, and LlaSMol, formulate molecular tasks as sequence-to-sequence or instruction-following problems, gaining task versatility but sacrificing explicit access to molecular topology and geometry [5; 29; 20; 27]. Multimodal molecular LLMs and molecule-text alignment models attempt to connect language representations with molecular graphs, 3D structures, or biochemical knowledge, improving molecule-text retrieval, captioning, and open-ended molecular question answering [14; 15; 13; 16; 12; 6]. Closely related efforts in materials science, such as MatterChat, align crystal-structure embeddings from a pretrained interatomic-potential model with an LLM for inorganic material property prediction and material question answering [23]. In contrast, SciCore-Mol targets molecular chemistry rather than crystalline materials and combines multiple pluggable cognition modules—geometry-aware molecular perception, latent molecular generation, and reaction-level quantitative reasoning—within one hidden-state interface. Another line of work augments LLMs with external tools or scientific agents. For example, ChemCrow and Coscientist use LLMs to plan chemical workflows and invoke expert-designed tools, while ChatMol and ChemAgent support molecular discovery and chemical reasoning through natural-language interaction [4; 3; 28; 22].

Despite these advances, many existing systems still rely on loose coupling between the LLM and specialized scientific computation. In such designs, the LLM mainly functions as a language interface, planner, decoder, or conditioning signal, whereas graph encoders, reaction predictors, diffusion generators, retrieval systems, or laboratory tools perform the core scientific operations. This interface improves performance on predefined tasks, but limits flexibility and compositional generalization: external scientific modules cannot directly interact with the LLM's internal reasoning state, and intermediate molecular, geometric, or numerical information must often be compressed back into text before being used by the LLM. This text-level interface may introduce information loss and reasoning bottlenecks, especially for tasks requiring structure-grounded perception, continuous molecular generation, or numerically sensitive reaction reasoning.

Our approach. Motivated by the limitations of both text-only molecular LLMs and loosely coupled tool-augmented systems, we design SciCore-Mol around a tighter integration principle: the LLM backbone remains a general-purpose reasoning and coordination model, while molecular cognition is provided by three pluggable specialist modules that are integrated at the hidden-state level rather than connected through text-only tool feedback. Three complementary modules address the three key limitations of text-only modeling:

1. **Geometry:** a GVP encoder [10] provides SE(3)-equivariant 3D molecular representations injected as virtual tokens into the LLM context.
2. **Generation:** a latent diffusion decoder conditioned on LLM hidden states generates structurally valid molecules in a continuous latent space, bypassing autoregressive SMILES decoding.

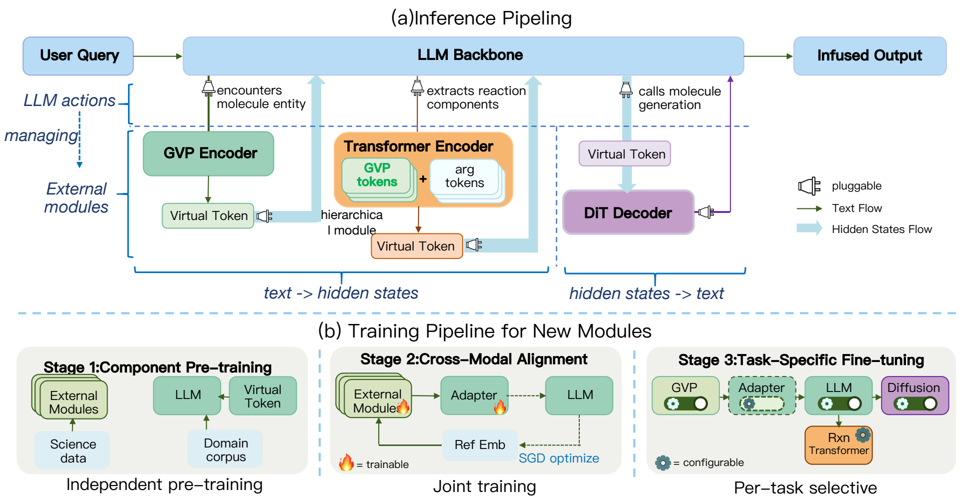


Figure 1. **(a)** Inference pipeline of SciCore-Mol. The LLM backbone selectively activates specialist modules based on task context. **(b)** Three-stage progressive training schedule.

3. **Reactions:** a Reaction Transformer processes multi-channel molecular tokens—combining structure, stoichiometric quantities, and functional roles—for robust inter-molecular reasoning.

Module dispatch is controlled by the LLM itself through predefined special tokens, enabling context-aware, on-demand invocation without external routing logic.

Summary of contributions.

- We present SciCore-Mol, a unified framework for augmenting LLMs with pluggable molecular cognition modules that integrate at the hidden-state level.
- We describe a progressive three-stage training recipe that stabilizes joint optimization across heterogeneous module objectives.
- We provide benchmark results on five evaluation suites (SMolInstruct, MMLU-Chemistry, ChemBench4K, ORD, MoleculeNet), confirming improvements in generation validity, reaction reasoning, and property prediction.

2 Approach

In this section we describe the SciCore-Mol architecture, training data, and progressive training pipeline.

2.1 Architecture

SciCore-Mol consists of a Qwen3-8B LLM backbone [26] with hidden dimension $d = 3{,}072$ and three specialist modules connected through learned projection layers. Figure 1 illustrates the inference pipeline.

2.1.1 Problem Formulation

An input instance is a token sequence $\mathbf{X} = (x_1, x_2, \ldots, x_n)$ interleaving natural-language tokens with molecular entity spans. A chemical entity detector, implemented as a token-level classifier combined with rule-based named entity recognition, identifies molecular spans and delimits them with special tokens $\langle\texttt{mol}\rangle \cdots \langle/\texttt{mol}\rangle$. Let $\mathbf{H}_\mathbf{X} \in \mathbb{R}^{n\times d}$ denote the hidden-state sequence produced by the LLM backbone from the textual input.

For each detected molecular span indexed by k, we construct a 3D molecular graph $\mathcal{G}_k$ from the canonicalized SMILES using RDKit, encode it with the Topological Perception Module, and project it into the LLM hidden space:

$$\mathbf{h}_k^{\mathrm{mol}} = f_{\mathrm{adapt}}\left(\mathrm{GVP}(\mathcal{G}_k)\right) \in \mathbb{R}^d, \quad k = 1, \ldots, K, \tag{1}$$

where f_{adapt} is a learnable MLP adapter and K is the number of detected molecular entities. The augmented hidden-state sequence is formed by concatenating the textual hidden states and the virtual structural tokens:

$$\tilde{\mathbf{H}}_{\mathbf{X}} = \mathrm{Concat}\left(\mathbf{H}_{\mathbf{X}}, \mathbf{h}_1^{\mathrm{mol}}, \ldots, \mathbf{h}_K^{\mathrm{mol}}\right). \tag{2}$$

The attention mask is extended accordingly so that subsequent generated tokens can attend to both textual tokens and virtual structural tokens. The LLM then generates output tokens autoregressively:

$$\mathbf{o}_t = \mathrm{LLM}_\theta\left(y_{<t}; \tilde{\mathbf{H}}_{\mathbf{X}}\right), \qquad p_\theta\left(y_t \mid y_{<t}, \mathbf{X}, \tilde{\mathbf{H}}_{\mathbf{X}}\right) = \mathrm{Softmax}\left(W_o \mathbf{o}_t\right)_{y_t}. \tag{3}$$

Beyond structural perception, SciCore-Mol selectively activates two additional modules based on task context. When a molecular generation task is identified, the LLM hidden states are projected as the condition for a latent diffusion decoder:

$$\mathbf{c} = \mathrm{TextProj}\left(\mathbf{H}_{\mathrm{LLM}}\right), \qquad \hat{\mathbf{z}}_0 = \mathrm{Sample}_{\epsilon_\theta}\left(\mathbf{z}_T; \mathbf{c}\right), \qquad \hat{s} = \mathrm{Dec}\left(\hat{\mathbf{z}}_0\right), \quad \mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{4}$$

Here $\mathrm{Sample}_{\epsilon_\theta}$ denotes the iterative DDPM/DDIM denoising process parameterized by the DiT denoiser ϵ_θ, rather than a single forward pass.

When a reaction-level query is encountered, the Reaction Transformer processes a sequence of composite molecular tokens:

$$\mathbf{r}_j = f_{\mathrm{mol}}\left(\mathbf{h}_j^{\mathrm{geo}}\right) + f_{\mathrm{amt}}\left(\bar{\mathbf{a}}_j\right) + \mathbf{e}^{\mathrm{role}}\left(\rho_j\right) + \mathbf{e}^{\mathrm{type}}\left(\tau_j\right), \quad j = 1, \ldots, J. \tag{5}$$

Here $\mathbf{h}_j^{\mathrm{geo}} \in \mathbb{R}^{256}$ is the GVP graph-level representation of molecule j, $\bar{\mathbf{a}}_j \in \mathbb{R}^{10}$ is an amount feature vector containing normalized stoichiometric quantities, missing-value masks, and auxiliary numerical indicators, ρ_j denotes the molecule's functional role, and τ_j distinguishes input molecules from target or masked molecules.

All three branches share the same LLM backbone. The Topological Perception Module provides molecular geometry, the Molecular Generation Module provides continuous structure generation, and the Reaction Sensing Module provides reaction-level quantitative reasoning. Each module can be independently enabled or disabled, yielding a plug-and-play architecture whose deployment cost scales with the required capabilities.

2.1.2 Topological Perception Module

SMILES strings encode molecular connectivity but do not explicitly represent 3D molecular geometry. We employ a GVP (Geometric Vector Perceptron) network [10] to encode RDKit-generated 3D conformers as geometry-aware molecular representations.

GVP Encoder. Given a molecular graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, each node v carries scalar features s_v, such as atomic number, formal charge, aromaticity, ring membership, and hydrogen count, and vector features $\mathbf{V}_v \in \mathbb{R}^{\nu \times 3}$ describing geometric directions. Each edge carries scalar features, such as interatomic distance, and vector features, such as unit direction. A simplified GVP layer transforms scalar-vector features as:

$$\mathbf{V}^h = W_V \mathbf{V}, \tag{6}$$

$$\mathbf{s}' = \sigma_s\left(W_s\left[\mathbf{s} \,\|\, \left\|\mathbf{V}^h\right\|_2\right] + \mathbf{b}_s\right), \tag{7}$$

$$\mathbf{V}' = g(\mathbf{s}') \odot \mathbf{V}^h. \tag{8}$$

No additive bias is applied to vector channels, so that the vector features preserve equivariance under rigid rotations. The scalar channel uses vector norms and is therefore invariant to rotations. We stack four GVPConv message-passing layers with hidden dimensions $(256, 16)$, followed by global pooling over node representations to obtain a graph-level representation $\mathbf{h}_{\text{geo}} \in \mathbb{R}^{256}$.

MLP Adapter. A two-layer MLP projects the graph-level geometric representation into the LLM hidden space:

$$\mathbf{h}_{\text{mol}} = W_2 \, \text{ReLU} \left(W_1 \mathbf{h}_{\text{geo}} + \mathbf{b}_1 \right) + \mathbf{b}_2 \in \mathbb{R}^d, \tag{9}$$

with dimensions $256 \to 2{,}048 \to 3{,}072$. The projected $\mathbf{h}_{\text{mol}}$ is used as a Virtual Structural Token, enabling subsequent self-attention layers to access 3D geometric information alongside textual context.

Chemical Entity Detection. A dedicated detector scans the input and identifies molecular entities. When an entity is detected, a *virtual step* is triggered: the detected molecular string is canonicalized, converted to a 3D conformer, encoded by the GVP encoder, projected by the adapter, and appended to the LLM's hidden-state context as a virtual structural token. This allows the autoregressive generation process to continue with geometry-aware molecular context, without requiring all molecular features to be pre-computed before decoding.

Takeaway

1. The GVP encoder preserves equivariance in vector features and produces rotation-invariant scalar summaries for molecular conditioning.
2. Virtual structural tokens provide geometry-aware grounding for the LLM without modifying the backbone architecture.

2.1.3 Molecular Generation Module

Autoregressive SMILES generation with generic language models remains challenging, as standard tokenizers are not explicitly aligned with SMILES grammar, and small token-level errors may lead to invalid molecules. We therefore integrate a latent diffusion module operating in a continuous molecular latent space.

SMILES Autoencoder. We employ a pre-trained BERT-style SMILES autoencoder to map molecules into compact latent representations $\mathbf{z}_0 \in \mathbb{R}^{127 \times 64}$. The autoencoder is trained with a shared BPE tokenizer of vocabulary size 300 and remains frozen during subsequent training. This design provides a stable continuous latent space that preserves molecular semantics and supports smooth interpolation.

Latent Diffusion. To model the distribution of molecular latent representations, we adopt a denoising diffusion probabilistic model in latent space. The forward diffusion process gradually perturbs $\mathbf{z}_0$ by injecting Gaussian noise over T timesteps:

$$q\left(\mathbf{z}_t \mid \mathbf{z}_{t-1}\right) = \mathcal{N}\left(\mathbf{z}_t; \sqrt{1-\beta_t}\, \mathbf{z}_{t-1}, \beta_t \mathbf{I}\right), \qquad t = 1, \ldots, T. \tag{10}$$

This Markov process admits the closed-form marginal

$$q\left(\mathbf{z}_t \mid \mathbf{z}_0\right) = \mathcal{N}\left(\mathbf{z}_t; \sqrt{\bar{\alpha}_t}\, \mathbf{z}_0, (1-\bar{\alpha}_t)\mathbf{I}\right), \tag{11}$$

where

$$\alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s. \tag{12}$$

To incorporate high-level semantic information, we extract hidden representations from the LLM backbone and project them into the diffusion conditioning space:

$$\mathbf{c} = \mathrm{TextProj}\left(\mathbf{H}_{\mathrm{LLM}}\right). \tag{13}$$

The reverse process is parameterized by a Transformer denoiser that predicts the injected noise conditioned on the noisy latent, diffusion timestep, and semantic condition:

$$\hat{\boldsymbol{\epsilon}} = \epsilon_\theta\left(\mathbf{z}_t, t, \mathbf{c}\right). \tag{14}$$

During inference, we typically start from Gaussian noise $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. For reference-guided generation, given a source molecule x_{src}, we initialize from its noised latent representation:

$$\mathbf{z}_{\mathrm{src}} = \mathrm{Enc}\left(x_{\mathrm{src}}\right), \qquad \mathbf{z}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{z}_{\mathrm{src}} + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \tag{15}$$

which enables structure-preserving molecular editing under condition $\mathbf{c}$. To strengthen conditional alignment, we further adopt classifier-free guidance during inference:

$$\hat{\boldsymbol{\epsilon}}_{\mathrm{cfg}} = \epsilon_\theta\left(\mathbf{z}_t, t, \varnothing\right) + s \cdot \left(\epsilon_\theta\left(\mathbf{z}_t, t, \mathbf{c}\right) - \epsilon_\theta\left(\mathbf{z}_t, t, \varnothing\right)\right), \tag{16}$$

where s is the guidance scale. The final denoised latent is decoded back into a SMILES sequence through the frozen autoencoder.

Takeaway

1. Continuous latent diffusion decouples molecular validity from token-level decoding constraints in the LLM.
2. Latent diffusion supports controllable reference-aware generation while preserving sufficient diversity for molecular exploration.

2.1.4 *Molecular Interaction Module*

Single-molecule tasks are handled by the Topological Perception Module, but inter-molecular reaction tasks—especially yield estimation—require modeling stoichiometric relationships across multiple participants. We design a Reaction Transformer in which each input token represents one molecule through the multi-channel composite encoding defined above. Figure 2 illustrates this token construction. Specifically, $\mathbf{h}_j^{\mathrm{geo}}$ is the GVP graph-level embedding of molecule j, $\bar{\mathbf{a}}_j \in \mathbb{R}^{10}$ contains normalized stoichiometric features and missing-value indicators, $\mathbf{e}^{\mathrm{role}}(\rho_j)$ is a learned role embedding for reactant, reagent, solvent, catalyst, product, and other reaction roles, and $\mathbf{e}^{\mathrm{type}}(\tau_j)$ distinguishes observed input molecules from masked or target molecules.

Architecture. The Reaction Transformer consists of a molecule projection layer ($256 \to 512$), an amount projection MLP ($10 \to 512$), learned role embeddings (11×512), type embeddings (2×512), a learnable `[CLS]` token, and a 6-layer Transformer encoder with 8 attention heads, hidden dimension 512, feedforward dimension 2048, GELU activation, and pre-LayerNorm. Missing channels are masked before projection, allowing the same architecture to handle product prediction, retrosynthesis, and yield estimation.

Task-Specific Heads. Four heads produce task outputs:

1. **Embedding reconstruction**: a linear projection ($512 \to 256$) predicts GVP embeddings of masked molecules and is trained with an InfoNCE loss with temperature $\tau = 0.07$.
2. **Amount prediction**: a linear projection ($512 \to 3$) predicts log-scale moles, mass, and volume, trained with Smooth L1 loss.
3. **Yield classification**: a linear projection from `[CLS]` ($512 \to 10$) classifies yield into 10 bins.
4. **Yield regression**: a linear projection from `[CLS]` ($512 \to 1$) predicts continuous yield,

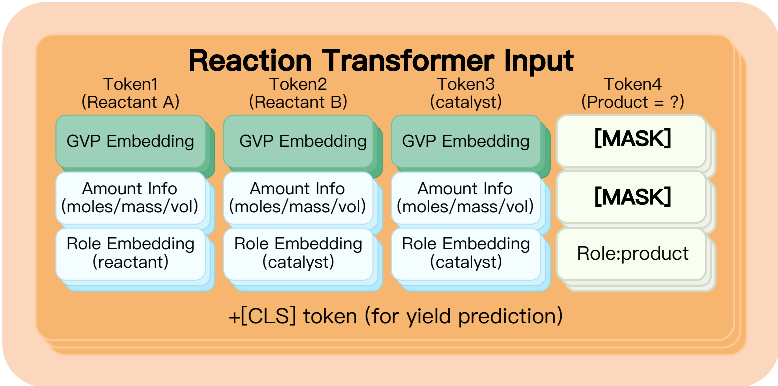


Figure 2. Reaction token construction in the Molecular Interaction Module. Each token combines molecular geometry embedding, amount features, and role/type signals, while masked targets and a `[CLS]` token support product and yield prediction.

trained with mean squared error.

The dual yield head combines coarse classification with continuous regression, where classification acts as a regularizer. Predicted molecular embeddings and yield values are fed back to the LLM as additional context for continued natural-language reasoning.

Takeaway

1. Multi-channel composite tokens combine geometry, stoichiometry, and functional role in a single reaction-level representation.
2. Masked channels allow a single Reaction Transformer to unify product prediction, retrosynthesis, amount prediction, and yield estimation.

2.2 Training Data

2.2.1 Continual Pre-training Corpus

The LLM backbone is continually pre-trained on a 400M-token chemistry-enriched corpus sampled from a 2.17B-token pool at a 1:1 domain-to-general ratio. Table 1 summarizes the composition.

Table 1. CPT corpus composition ($\sim$400M tokens sampled from $\sim$2.17B pool).

Source	**Type**	**Pool size**	**Sampled**
Chemical literature	Domain	283M	$\sim$52M
ORD [11]	Domain	1.27B	$\sim$148M
Nemotron-CC [18]	General	1.5B	$\sim$50M
Nemotron-CC-Math	General	1.5B	$\sim$50M
Nemotron-Code-Synthetic	General	1.5B	$\sim$50M
Nemotron-SFT	General	1.5B	$\sim$50M

A KL-divergence regularization term against the frozen Qwen3-8B base model is applied during continual pre-training:

$$\mathcal{L}_{\mathrm{KL}} = D_{\mathrm{KL}}\left(p_\theta \,\|\, p_{\mathrm{ref}}\right), \tag{17}$$

where p_θ is the output distribution of the continually trained model and p_{ref} is the output distribution of the frozen reference model.

2.2.2 *Instruction Fine-tuning Data*

We construct 300K instruction–response pairs ($\sim$245M tokens) at a 2.5:1 general-to-domain ratio. The general portion, consisting of 214,285 pairs or 71.4%, comes from Nemotron-Post-Training-v2 [18]. The chemistry portion, consisting of 85,715 pairs or 28.6%, is sourced from SMolInstruct [19]:

- **Reaction synthesis** (36,083 samples): forward synthesis (18,144) and retrosynthesis (17,939).
- **Name conversion** (38,995 samples): SMILES-to-formula (10,094), IUPAC-to-formula (9,978), SMILES-to-IUPAC (9,630), and IUPAC-to-SMILES (9,293).
- **Molecule generation** (4,240 samples): text-conditioned de novo design, routed to the diffusion decoder.
- **Molecule captioning** (4,162 samples): natural-language descriptions of molecular structures.
- **Property prediction** (2,235 samples): physicochemical property tasks from MoleculeNet [25].

Task balancing uses temperature-based allocation with $\alpha = 0.5$. Binary classification tasks, including BBBP, ClinTox, and HIV, are further balanced by Yes/No label. All samples are filtered to at most 2,048 tokens per conversation. Cross-file deduplication is performed using SHA-1 fingerprints of normalized message sequences.

2.3 Progressive Training Strategy

Training all modules simultaneously from scratch leads to unstable optimization because the LLM, geometric encoder, diffusion decoder, and reaction module have heterogeneous objectives and different convergence speeds. We therefore adopt a progressive three-stage strategy.

2.3.1 *Stage I: Molecular Foundation Pre-training*

Step 1: Component Pre-training. All components are pre-trained independently. The LLM backbone undergoes KL-regularized continual pre-training on the 400M-token corpus described above. The GVP encoder is pre-trained on molecular property prediction tasks to learn meaningful geometric representations. The diffusion decoder is pre-trained on large-scale molecular databases to establish a continuous latent space for molecular generation. The Reaction Transformer is pre-trained on ORD data covering product prediction, retrosynthesis, amount prediction, and yield estimation. The chemical entity recognizer is trained separately on annotated chemical text corpora.

Step 2: Cross-Modal Alignment. After component pre-training, the LLM backbone is frozen and only the GVP encoder and MLP adapter receive gradients. For each molecule, the adapter output $\mathbf{h}_{\mathrm{mol}}$ and the LLM text embedding $\mathbf{h}_{\mathrm{text}}$ of its natural-language description form a positive pair. We use a symmetric contrastive objective:

$$\mathcal{L}_{\mathrm{m2t}} = -\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \log \frac{\exp\left(\mathrm{sim}\left(\mathbf{h}^{i}_{\mathrm{mol}}, \mathbf{h}^{i}_{\mathrm{text}}\right)/\tau\right)}{\sum_{j\in\mathcal{B}} \exp\left(\mathrm{sim}\left(\mathbf{h}^{i}_{\mathrm{mol}}, \mathbf{h}^{j}_{\mathrm{text}}\right)/\tau\right)}, \tag{18}$$

$$\mathcal{L}_{\mathrm{t2m}} = -\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \log \frac{\exp\left(\mathrm{sim}\left(\mathbf{h}^{i}_{\mathrm{text}}, \mathbf{h}^{i}_{\mathrm{mol}}\right)/\tau\right)}{\sum_{j\in\mathcal{B}} \exp\left(\mathrm{sim}\left(\mathbf{h}^{i}_{\mathrm{text}}, \mathbf{h}^{j}_{\mathrm{mol}}\right)/\tau\right)}, \tag{19}$$

$$\mathcal{L}_{\mathrm{align}} = \frac{1}{2}\left(\mathcal{L}_{\mathrm{m2t}} + \mathcal{L}_{\mathrm{t2m}}\right). \tag{20}$$

The alignment objective is trained on approximately 100K SMILES-description pairs from the ChatMol corpus [28]. By keeping the LLM frozen, the adapter learns to map geometric representations into the LLM's existing hidden space without disrupting its language modeling capability.

2.3.2 *Stage II: Joint Multi-Task Training*

With the modules aligned, the full system—including the GVP encoder, adapter, LLM, diffusion decoder, and Reaction Transformer—is jointly trained on the 300K SFT corpus. The training tasks include name conversion, property prediction, molecule generation, forward synthesis, retrosynthesis, captioning, reaction product prediction, yield estimation, and stoichiometric reasoning. The diffusion module is simultaneously trained on text-to-structure generation and bridge-diffusion editing objectives, while the Reaction Transformer is optimized with embedding reconstruction, amount prediction, and yield prediction objectives.

2.3.3 Stage III: Task-Specific Fine-tuning

Individual modules can be selectively frozen or unfrozen for downstream deployment. For example, freezing the diffusion decoder and Reaction Transformer for property-prediction-only deployment reduces inference cost, whereas freezing the GVP encoder and diffusion decoder allows task-specific adaptation to focus on reaction tasks.

2.4 Training Objectives

Language Modeling Loss. The language modeling objective is:

$$\mathcal{L}_{\mathrm{LM}} = -\sum_{t=1}^{T_y} \log p_\theta \left(y_t \mid y_{<t}, \mathbf{X}, \tilde{\mathbf{H}}_{\mathbf{X}} \right). \tag{21}$$

Latent Diffusion Loss. Given a clean latent representation $\mathbf{z}_0$, a diffusion timestep t, and Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we first construct the noisy latent

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{I}\right), \tag{22}$$

and train the denoiser to predict the injected noise conditioned on the semantic signal $\mathbf{c}$:

$$\mathcal{L}_{\mathrm{diff}} = \mathbb{E}_{\mathbf{z}_0, \boldsymbol{\epsilon}, t} \left[\|\boldsymbol{\epsilon} - \epsilon_\theta \left(\mathbf{z}_t, t, \mathbf{c} \right)\|_2^2 \right]. \tag{23}$$

Reaction Loss. The Reaction Transformer is trained with a combination of embedding reconstruction, amount prediction, and yield prediction losses:

$$\mathcal{L}_{\mathrm{rxn}} = \lambda_{\mathrm{emb}}\mathcal{L}_{\mathrm{emb}} + \lambda_{\mathrm{amt}}\mathcal{L}_{\mathrm{amt}} + \lambda_{\mathrm{yield}}\mathcal{L}_{\mathrm{yield}}. \tag{24}$$

The amount prediction loss is a Smooth L1 loss:

$$\mathcal{L}_{\mathrm{amt}} = \mathrm{SmoothL1}\left(\hat{\mathbf{a}}, \mathbf{a}\right). \tag{25}$$

The yield prediction loss combines regression and classification:

$$\mathcal{L}_{\mathrm{yield}} = \lambda_{\mathrm{reg}} \left\|\hat{y}_{\mathrm{reg}} - y\right\|_2^2 + \lambda_{\mathrm{cls}}\, \mathrm{CE}\left(\hat{\mathbf{y}}_{\mathrm{cls}}, \mathrm{bin}(y)\right). \tag{26}$$

Stage-wise Optimization. During Stage I, components are optimized with their corresponding objectives rather than as a single fully coupled system:

$$\mathcal{L}_{\mathrm{I}}^{\mathrm{LLM}} = \mathcal{L}_{\mathrm{LM}} + \lambda_{\mathrm{KL}} D_{\mathrm{KL}}\left(p_\theta \,\|\, p_{\mathrm{ref}}\right), \tag{27}$$

$$\mathcal{L}_{\mathrm{I}}^{\mathrm{align}} = \mathcal{L}_{\mathrm{align}}, \tag{28}$$

$$\mathcal{L}_{\mathrm{I}}^{\mathrm{diff}} = \mathcal{L}_{\mathrm{diff}}, \tag{29}$$

$$\mathcal{L}_{\mathrm{I}}^{\mathrm{rxn}} = \mathcal{L}_{\mathrm{rxn}}. \tag{30}$$

During Stage II, the full model is jointly optimized with:

$$\mathcal{L}_{\mathrm{II}} = \lambda_{\mathrm{LM}}\mathcal{L}_{\mathrm{LM}} + \lambda_{\mathrm{align}}\mathcal{L}_{\mathrm{align}} + \lambda_{\mathrm{diff}}\mathcal{L}_{\mathrm{diff}} + \lambda_{\mathrm{rxn}}\mathcal{L}_{\mathrm{rxn}}. \tag{31}$$

When a mini-batch does not contain a certain task type, the corresponding loss term is masked out. Stage III reuses the same objective family but selectively freezes or unfreezes modules depending on the downstream task.

All stages use AdamW with a cosine learning rate schedule and linear warmup. Stage I uses fp16 mixed precision, while Stage II uses bf16 to improve numerical stability for fine-grained module parameters. Training is conducted on NVIDIA A800 80GB GPUs.

Takeaway

1. Stage-wise training decouples heterogeneous objectives during early optimization and reduces gradient conflicts.
2. Joint training aligns the LLM, GVP encoder, diffusion decoder, and Reaction Transformer within a shared hidden-state interface.
3. Selective module freezing at Stage III makes the system cost-adaptive for different downstream deployment scenarios.

3 Experiments

3.1 Experimental Setup

Baselines. We compare SciCore-Mol (Qwen3-8B backbone, ∼8.6B total parameters) against:

- **GPT-4o / GPT-5** [8; 21]: closed-source frontier models. Process molecular inputs as raw SMILES text.
- **Intern-S1-mini** [2]: science-specialized reasoning model.
- **LlaSMol-Mistral-7B** [27]: chemistry-tuned model on SMolInstruct, text-centric paradigm.
- **Qwen3-8B** [26]: the official 8B-parameter Qwen3 base model without chemistry-domain adaptation, establishing a lower bound.
- **Qwen3-8B-Chem**: our chemistry-adapted backbone obtained from Qwen3-8B through chemistry-domain continued pretraining and supervised instruction tuning, without pluggable modules. This baseline isolates domain adaptation from module contributions.

Evaluation suites.

- **SMolInstruct** [27]: generative benchmark spanning six molecular task families.
- **MMLU-Chemistry** [7]: five chemistry subjects from MMLU, testing general knowledge retention.
- **ChemBench4K** [17]: multiple-choice chemistry benchmark covering synthesis, captioning, and yield tasks.
- **ORD** [11]: reaction-centered benchmark covering product prediction, yield estimation, and retrosynthesis.
- **MoleculeNet** [25]: molecular property prediction benchmark (classification and regression).

Metrics. Validity (%) and RDK fingerprint Tanimoto similarity (RDK-FTS) for structure generation/translation; F1 for classification; MAE/RMSE for regression; METEOR for captioning; accuracy and nDCG for multiple-choice/ranking tasks.

3.2 Main Comparison

Figure 3 presents capability radar charts across five dimensions aggregated from all benchmarks.

SciCore-Mol achieves a balanced profile across all five dimensions. Notably:

- **Molecule Generation and Synthesis Reasoning** benefit most from pluggable modules; both dimensions show clear gains over the text-only Qwen3-8B-Chem baseline.
- **Knowledge Core** (MMLU-Chemistry) remains competitive, confirming that KL-regularized CPT and progressive training preserve general scientific reasoning.
- Closed-source models (GPT-5) lead on Knowledge Core due to scale, but SciCore-Mol is competitive on structure and reaction dimensions at a fraction of the parameter count.

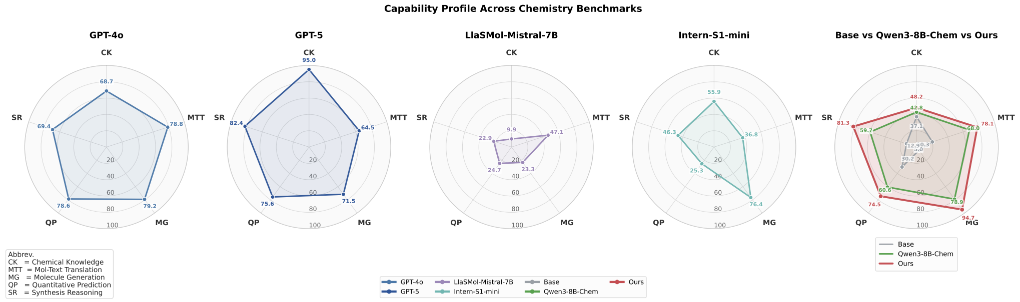


Figure 3. Per-model capability radar across five dimensions: Knowledge Core, Mol-Text Translation, Molecule Generation, Quantitative Prediction, and Synthesis Reasoning. Scores normalized to 0–100.

Table 2. Supplementary: GVP ablation on MoleculeNet property tasks. Higher is better for F1 (↑); lower is better for MAE (↓).

Model	MoleculeNet Classification (F1)↑					MoleculeNet Regression (MAE)↓			
	BBBP	**Tox21**	**ClinTox**	**HIV**	**BACE**	**ESOL**	**FreeSolv**	**Lipo**	**QM9**
Qwen3-8B	0.47	0.27	0.50	0.25	0.61	1.11	0.72	1.00	0.92
Qwen3-8B-Chem-PT	0.53	0.35	0.39	0.56	0.63	0.98	0.71	0.92	0.86
Qwen3-8B-Chem	0.51	0.53	0.50	0.70	0.60	0.93	0.97	0.89	0.90
Qwen3-8B-Chem-PT+GVP	0.49	0.85	0.53	0.94	0.65	0.88	0.81	0.86	0.78

3.3 Module Ablation Studies

3.3.1 Topological Perception Module (GVP)

Table 2 ablates the GVP encoder on MoleculeNet property prediction. Four variants are compared: the Qwen3-8B base model, a continued-pretraining-only chemistry variant, the instruction-tuned Qwen3-8B-Chem backbone, and the GVP-augmented chemistry backbone.

Adding the GVP encoder consistently improves both classification (F1) and regression (MAE) tasks across all evaluated datasets. On Tox21, HIV, and BACE, the improvement from Qwen3-8B-Chem-PT to Qwen3-8B-Chem-PT+GVP is particularly large, confirming that these tasks are sensitive to geometric features that text-only models cannot capture.

Takeaway

1. GVP provides complementary geometric signal unavailable from SMILES text representations.
2. Classification and regression both benefit, suggesting broad utility across property types.

3.3.2 Molecular Generation Module (Diffusion)

Table 3 compares average main reward and structural similarity (RDK-FTS) to source molecular on a drug optimization task across model variants.

Our model achieves a high average reward (0.2380), which is competitive with GPT-4o (0.2545) and surpasses the fine-tuned Intern-S1-mini (0.2085).Structural similarity is lower for the diffusion model, reflecting that it optimizes property objectives rather than scaffold preservation by defaulta tradeoff controllable through diffusion for editing tasks.

Takeaway

1. Latent diffusion generation outperforms autoregressive SMILES decoding on reward-oriented tasks.
2. Diffusion enables scaffold-preserving optimization when structural similarity is required.

3.3.3 Reaction Sensing Module (Reaction Transformer)

On the ORD benchmark, removing the Reaction Transformer causes severe performance degradation:

- Product prediction validity: 42.0% → 11.4%
- Product + yield joint validity: 97.6% → 11.5%

Table 3. Comparison of average main reward and structure similarity (RDK-FTS) across different model variants. Higher is better.

Model	Avg. Main Reward	RDK-FTS
GPT-4o	0.2545	0.7276
GPT-5	0.2508	0.7436
Intern-S1-mini	0.2097	0.7263
Intern-S1-mini (SFT)	0.2085	0.8625
Qwen3-8B	0.1465	0.8795
Qwen3-8B-Chem	0.2202	0.3823
SciCore-Mol	0.2380	0.5442

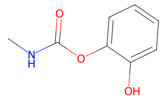
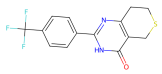
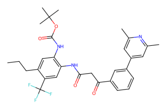
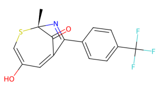
	(a) Molecular Generation	(b) Product + Yield Prediction	(c) Molecular Explanation
Input	**Q:** Generate a tankyrase-inhibitor-like thiopyranopyrimidine with a para-`CF3` phenyl substituent.	**Q:** Given two reactants, predict the coupled product and yield.	**Q:** Provide a brief overview of this molecule.  Input molecule: aromatic carbamate.
Ground Truth	 Reference: thiopyranopyrimidine + para-`CF3` phenyl.	 Reference product; yield **76.00%**.	**A:** Carbamate pesticide derived from carbamic acid; used in agricultural or residential contexts.
Baseline	**GPT-5:** similarity 0.537. **Intern-S1-mini:** 0.043.	**GPT-4o/GPT-5:** separated reactants or incomplete product sets. **Qwen3-8B-Chem:** unrelated aryl ketone-like product.	**GPT-4o/Qwen3-8B:** confuse the molecule with a simple aromatic ester or plasticizer.
SciCore-Mol	 **P:** similarity 0.607; preserves more scaffold-level information.	**P:** coupled-product pattern recovered; predicted yield **57.60%** (score 0.819).	**P:** selects the carbamate-pesticide description and grounds the decision in the carbamate motif.

Table 4. Representative case studies across the three inference pathways. Molecule renderings replace long SMILES strings, and text colors highlight the question, ground truth, incorrect baseline behavior, and SciCore-Mol predictions.

- Reactant prediction validity: 74.7% $\rightarrow$ 24.6%
- Yield MAE: 0.34 $\rightarrow$ 0.52

These results confirm that reaction-level reasoning cannot be achieved by the LLM backbone alone, even after domain pre-training. The multi-channel composite token encoding—combining GVP structure, stoichiometric quantities, and role labels—provides the structured input necessary for reliable product and yield prediction.

Takeaway

1. A dedicated reaction module is necessary for reaction-level quantitative reasoning.
2. The multi-channel molecular token encoding handles diverse reaction task types under a unified architecture.

4 Case Studies

We follow the three inference pathways introduced above and present representative cases in a compact qualitative format. Since long SMILES strings are difficult to read in the main text, molecule structures are represented by concise captions and replaceable structure panels.

Takeaway. These cases show that SciCore-Mol is useful beyond multiple-choice settings: it handles open-form molecule generation, reaction-level product/yield reasoning, and structure-aware molecular explanation. At the same time, the product-yield example shows a realistic limitation: the main product pattern is recovered, but the yield remains underestimated.

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